

Software Requirements Specification (SRS)

Non-Standard Format Data Extraction System & Verification Engine

Document Control

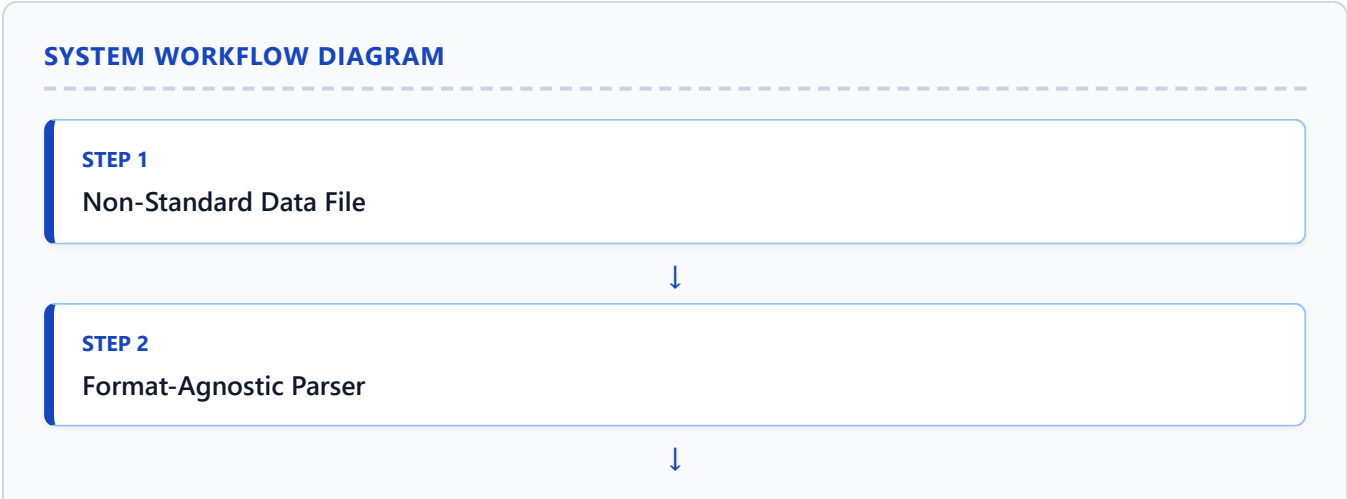
Item	Details
Document Title	SRS: System for Extracting Data from Non-Standard Formats & Performance Validation
System Module	Data Extraction & Verification Engine
Primary AI Engine	GPT Opus 120B Large Language Model
Document Version	1.0.0 (Pre-Implementation Baseline)
Target Audience	Software Architects, QA Lead Engineers, Data QC Testers

1. Executive Summary & Individual Flow Diagram

1.1 Purpose

Legacy plant maintenance logs are frequently stored in non-standard formats with irregular layout patterns, merged spreadsheet cells, informal column titles, and varying terminology. This specification defines the requirements for a resilient extraction system capable of interpreting non-standard inputs, normalizing raw content into standardized schemas, and validating extraction accuracy using defined verification test suites.

1.2 Individual Module Flow Diagram



STEP 3

GPT Opus 120B Model

**STEP 4**

Context-Aware Field Binding

**STEP 5**

Verification Test Suite

*Scenario 1: Clean Data / Scenario 2: Noisy Data / Scenario 3: Mixed Data***STEP 6**

100% Master Schema Precision

STEP 7

Noise Suppression & Intent Extraction

STEP 8

Multi-Format Cross Validation

**STEP 9**

Standardized Output Database

2. Theoretical Processing Logic

- **Format Agnostic Parsing:** The system parses input documents regardless of layout structure or custom column names.
- **Context-Aware Field Binding:** Using deep semantic understanding provided by **GPT Opus 120B**, non-standard column headers (e.g., "설비명칭", "Machine ID", "작업내용") are automatically mapped to standard target fields (`Equipment Name` , `Equipment Code` , `Report Content`).
- **Validation & Isolation:** Entries failing basic mandatory constraints are highlighted for user review or isolated in Quarantine.

3. Quality Control (QC) Verification Test Scenarios

Quality Control engineers MUST evaluate system performance across three specific verification scenarios:

3.1 Verification Scenario 1: Clean Data Benchmark

- **Input Specification:** Standardized files with clear column headers and complete entries.
- **Expected QC Outcome:**
 - Zero missing mandatory fields (Site , Process , Maintenance Part , Equipment Code , Equipment Name , Representative Work , Priority , Category).
 - 100% correct data type enforcement (e.g., ISO date formatting).

3.2 Verification Scenario 2: Noisy Data Benchmark

- **Input Specification:** Files containing OCR noise, placeholder tokens ("N/A", "확인중", "TBD"), text truncations, and informal slang.
- **Expected QC Outcome:**
 - Automatic suppression of meaningless noise tokens.
 - Accurate extraction of real underlying maintenance intent.
 - Flagging incomplete records without causing processing failures.

3.3 Verification Scenario 3: Mixed Data Benchmark

- **Input Specification:** Batch uploads combining spreadsheets, presentation decks, PDF invoices, image scans, and email messages.
- **Expected QC Outcome:**
 - Unified output dataset standardized to the master target schema.
 - Zero data loss or corruption during multi-format batch ingestion.

4. Operational Requirements

- **Accuracy Threshold:** Minimum 95% field extraction accuracy on noisy data inputs.
- **Fault Tolerance:** Malformed rows MUST be quarantined without breaking the pipeline.
- **Auditability:** All extraction actions MUST retain line-level source traceability.